

# PH102: Tutorial 10 solution

1.

$$\begin{aligned}
 \vec{M} &= a\hat{k} \left( \hat{k} \cdot \vec{r} \right)^2 \\
 (a) \quad \rho_m &= -\vec{\nabla} \cdot \vec{M} \\
 &= -\vec{\nabla} \cdot \left[ a\hat{k} \left( \hat{k} \cdot \vec{r} \right)^2 \right] \\
 &= -a\hat{k} \cdot \vec{\nabla} \left[ \left( \hat{k} \cdot \vec{r} \right)^2 \right] \\
 &= -2a \left( \hat{k} \cdot \vec{r} \right) \hat{k} \cdot \vec{\nabla} \left( \hat{k} \cdot \vec{r} \right) \\
 &= -2a \left( \hat{k} \cdot \vec{r} \right) \\
 &= -2ar \cos \theta
 \end{aligned}$$

(b)

$$\begin{aligned}
 \sigma_m &= \vec{M} \cdot \vec{r} = a \left( \hat{k} \cdot \hat{r} \right) \left( \hat{k} \cdot \vec{r} \right)^2 \\
 &= ar^2 \left( \hat{k} \cdot \hat{r} \right)^3 \\
 &= ar^2 \cos^3 \theta
 \end{aligned}$$

(c)

$$\begin{aligned}
 \vec{k}_b &= \vec{M} \times \hat{r} = a \left( \hat{k} \cdot \vec{r} \right)^2 \left( \hat{k} \times \hat{r} \right) \\
 &= \hat{\phi} a \left( \hat{k} \cdot \vec{r} \right) \sin \theta = \hat{\phi} ar \cos \theta \sin \theta
 \end{aligned}$$

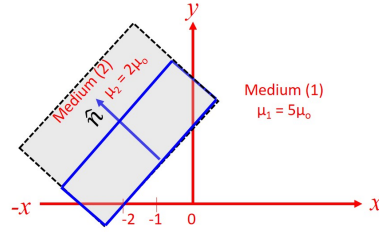
(d)

$$\begin{aligned}
 \vec{m} &= \int \vec{M} d\tau \\
 &= a \int \hat{k} \left( \hat{k} \cdot \vec{r} \right)^2 d\tau
 \end{aligned}$$

$$\begin{aligned}
&= 2\pi a \hat{k} \int_0^R \int_0^\pi r^4 \cos^2 \theta \sin \theta d\theta dr \\
&= \frac{4\pi}{3} a \hat{k} \int_0^R r^4 dr \\
&= \frac{4\pi a R^5}{15} \hat{k}
\end{aligned}$$

2.

(I) Since  $y - x - 2 = 0$  is a plane,  $y - x \leq 2$  or  $y \leq x + 2$  is the region as shown in the below figure. For instance  $(0,0)$  is in this region since  $0 - 0 - 2 < 0$ . Let's assume that plane is



described by the function  $f(x, y) = y - x - 2$ . Therefore, a unit vector normal to the plane is given by

$$\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{\hat{y} - \hat{x}}{\sqrt{2}}$$

(a)

$$\begin{aligned}
\mathbf{M}_1 &= \chi_{m1} H_1 = (\mu_{r1} - 1) \mathbf{H}_1 = (5 - 1)(-2\hat{i} + 6\hat{j} + 4\hat{k}) \\
&= -8\hat{i} + 24\hat{j} + 16\hat{k} A/m
\end{aligned}$$

$$\begin{aligned}
\mathbf{B}_1 &= \mu_1 \mathbf{H}_1 = \mu_0 \mu_{r1} \mathbf{H}_1 = 4\pi \times 10^{-7} (5)(-2\hat{i} + 6\hat{j} + 4\hat{k}) \\
&= -12.57\hat{i} + 37.7\hat{j} + 25.13\hat{k} \mu Wb/m^2
\end{aligned}$$

(b)

$$\begin{aligned}
\mathbf{H}_{1n} &= (\mathbf{H}_1 \cdot \hat{n}) \hat{n} \\
&= -4\hat{i} + 4\hat{j}
\end{aligned}$$

But we have,

$$\mathbf{H}_1 = \mathbf{H}_{1n} + \mathbf{H}_{1t}$$

Hence,

$$\mathbf{H}_{1t} = \mathbf{H}_1 - \mathbf{H}_{1n} = 2\hat{i} + 2\hat{j} + 4\hat{k}$$

Using the boundary conditions, we have

$$\mathbf{H}_{2t} = \mathbf{H}_{1t} = 2\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\mathbf{B}_{2n} = \mathbf{B}_{1n} \implies \mu_2 \mathbf{H}_{2n} = \mu_1 \mathbf{H}_{1n}$$

or

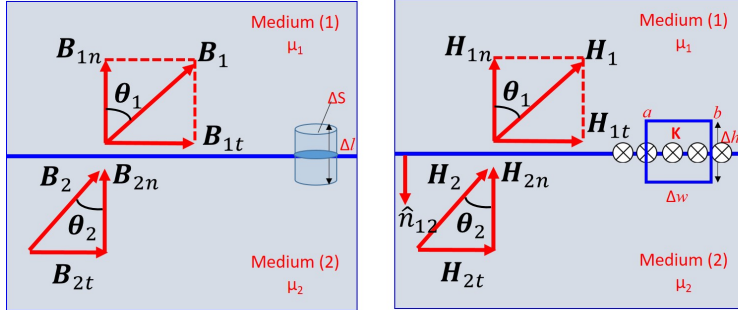
$$\mathbf{H}_{2n} = \frac{\mu_1}{\mu_2} \mathbf{H}_{1n} = \frac{5}{2}(-4\hat{i} + 4\hat{j}) = -10\hat{i} + 10\hat{j}$$

$$\mathbf{H}_2 = \mathbf{H}_{2n} + \mathbf{H}_{2t} = -8\hat{i} + 12\hat{j} + 4\hat{k} A/m$$

$$\mathbf{B}_2 = \mu_2 \mathbf{H}_2 = \mu_0 \mu_{r2} \mathbf{H}_2 = (4\pi \times 10^{-7})(2)(-8\hat{i} + 12\hat{j} + 4\hat{k})$$

$$= -20.11\hat{i} + 30.1\hat{j} + 10.05\hat{k} \mu Wb/m^2$$

(II) (a) From  $H_{1t} - H_{2t} = K$  and  $M = \chi_m H$ , we obtain:



$$\frac{M_{1t}}{\chi_{m1}} - \frac{M_{2t}}{\chi_{m2}} = K$$

Also we have  $B_{1n} - B_{2n} = K$  and  $B = \mu H = (\mu/\chi_m)M$ , we get:

$$\frac{\mu_1 M_{1n}}{\chi_{m1}} = \frac{\mu_2 M_{2n}}{\chi_{m2}}$$

(b) From  $B_1 \cos \theta_1 - B_{1n} = B_{2n} = B_2 \cos \theta_2$  (1) and

$$\frac{B_1 \sin \theta_1}{\mu_1} = H_{2t} = k + H_{2t} = k + \frac{B_2 \sin \theta_2}{\mu_2} \quad (2)$$

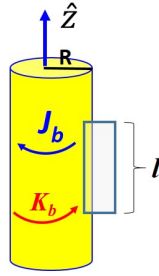
Dividing (2) by (1) we get

$$\frac{\tan \theta_1}{\mu_1} = \frac{k}{B_2 \cos \theta_2} + \frac{\tan \theta_2}{\mu_2} = \frac{\tan \theta_2}{\mu_2} \left( 1 + \frac{k\mu_2}{B_2 \sin \theta_2} \right)$$

Therefore we will have finally,

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2} \left( 1 + \frac{k\mu_2}{B_2 \sin \theta_2} \right)$$

3. (a)



$$\mathbf{M} = ks\hat{z}; \mathbf{J}_b = \nabla \times \mathbf{M} = -k\hat{\phi}; \mathbf{K}_b = \mathbf{M} \times \hat{n} = kR\hat{\phi}$$

$B$  is in the  $z$ -direction. This is essentially a superposition of solenoids. So  $B = 0$  outside. Using the amperian loop as shown (shaded) in the figure-inner side at radius  $s$  we have:

$$\oint \mathbf{B} \cdot d\mathbf{l} = Bl = \mu_0 I_{enc} = \mu_0 \left[ \int J_b da + K_b l \right] = \mu_0 [-kl(R-s) + kRl] = \mu_0 kls$$

Therefore,

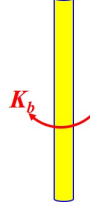
$$B = \mu_0 ks\hat{z} \quad \text{inside}$$

(b) By symmetry,  $\mathbf{H}$  points in the  $z$  direction. That same amperian loop gives  $\oint \mathbf{H} \cdot d\mathbf{l} = Hl = \mu_0 I_{fenc} = 0$ , as there is no free current here. So,  $H = 0$  and hence  $B = \mu_0 M$ . Outside  $\mathbf{M} = 0$ , so  $\mathbf{B} = 0$ ; inside  $M = ks\hat{z}$ , so  $B = \mu_0 ks\hat{z}$ .

4. (a) The field of a magnetized sphere is  $\frac{2}{3}\mu_0 M$  (As done in the class). So,  $B = B_0 - \frac{2}{3}\mu_0 M$  with the sphere removed. In the cavity,  $H = \frac{1}{\mu_0} B$ , so  $H = \frac{1}{\mu_0} \left( B_0 - \frac{2}{3}\mu_0 M \right) = H_0 + M - \frac{2}{3}M \implies$

$$H = H_0 + \frac{1}{3}M.$$

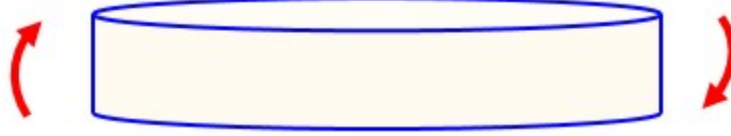
(b) The field inside a long solenoid is  $\mu_0 K$ , here  $K = M$ , so the field of the bound current on



the inside surface of the cavity is  $\mu_0 M$ , pointing down. Therefore,  $B = B_0 - \mu_0 M$ ;

$$H = \frac{1}{\mu_0} (B_0 - \mu_0 M) = \frac{1}{\mu_0} B_0 - M \implies H = H_0$$

(c) For this case the bound currents are small, and far away from the center, so  $B = B_0$ ,



$$\text{while } H = \frac{1}{\mu_0} B_0 = H_0 + M \implies H = H_0 + M$$

In the wafer,  $B$  is the field in the medium; in the needle,  $H$  is the  $H$  in the medium; in the sphere both  $B$  and  $H$  are modified.

5. Using the Amperes law we have

$$\oint H \cdot dl = H(2\pi s) = I_{fenc} = \begin{cases} I(s^2/a^2), & (s < a) \\ I, & (s > a) \end{cases}$$

Therefore,

$$H = \begin{cases} \frac{Is}{2\pi a^2}, & (s < a) \\ \frac{I}{2\pi s}, & (s > a) \end{cases}$$

So,

$$B = \mu H = \begin{cases} \frac{\mu_0(1 + \chi_m)Is}{2\pi a^2}, & (s < a) \\ \frac{\mu_0 I}{2\pi s}, & (s > a) \end{cases}$$

Further we know that

$$J_b = \chi_m J_f \quad \text{with} \quad J_f = \frac{I}{\pi a^2}$$

So

$$\boxed{J_b = \frac{\chi_m I}{\pi a^2}} \quad (\text{same direction as } I)$$

$$K_b = M \times \hat{n} = \chi_m H \times \hat{n} \implies \boxed{K_b = \frac{\chi_m I}{2\pi a}} \quad \text{opposite direction to } I$$

$$I_b = J_b(\pi a^2) + K_b(2\pi a) = \chi_m I - \chi_m I = 0 \quad (\text{which is quite expected also!})$$

6. (a)  $\vec{B}$  field produced by an infinite straight wire carrying current  $I$  at a point distance  $r$  is given by

$$B = \frac{\mu_0 I}{2\pi r}, \text{ its direction being } \perp \text{ to the wire.}$$

Thus the magnetic flux crossing the loop due to the wire further away from the loop is

$$\Phi_1 = \int_{2d}^{3d} \frac{\mu_0 I d}{2\pi r} dr = \frac{\mu_0 I d}{2\pi} \ln \frac{3}{2}$$

directing into the page. The other wire which is nearer to the loop, gives rise to the magnetic flux as

$$\Phi_2 = \int_d^{2d} \frac{\mu_0 I d}{2\pi r} dr = \frac{\mu_0 I d}{2\pi} \ln 2$$

pointing out from the page. Hence the total flux is

$$\Phi = \Phi_2 - \Phi_1 = \frac{\mu_0 I d}{2\pi} \ln \frac{4}{3}$$

pointing out from the page. The emf induced in the square loop is therefore,

$$\epsilon = -\frac{d\Phi}{dt} = -\frac{\mu_0 d}{2\pi} \ln \frac{4}{3} \frac{dI}{dt}$$

- (b) The magnetic field produced by the induced current tends to oppose the change of the magnetic flux, so that this field will direct into the page. Then by the right hand rule the induced current is clockwise.